Harry Nguyen

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Education

Georgia State University Expected May 2027
Bachelor of Science in Computer Science · **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

Work Experience

Google | Software Engineer Intern — Chrome Browser May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and cross-language compatibility
- Profiled settings navigation as p99 bottleneck at 1,200ms; designed **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven **TypeScript/React** observer system decoupling UI state propagation across **25K+ lines** of Chromium at **95% test coverage**; adopted into production branch by senior Chrome engineers

Develop for Good | Software Engineer Intern May 2024 – Aug 2024

- Built **Django REST** BaaS on **AWS** with JWT-based auth and auto-scaling — **500+ concurrent users**, **90% deployment time reduction** via GitHub Actions CI/CD
- Diagnosed N+1 query bottleneck degrading response to 3s+; redesigned with **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

CoderPush | Software Engineer Intern Sep 2023 – Dec 2023

- Designed idempotent **REST APIs** with **Redis** distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions; financial correctness under distributed failures
- Diagnosed **DynamoDB hot-partition failure** in high-throughput data pipeline at **9,000+ req/sec**; redesigned partition key strategy — **30% read throughput improvement** under sustained concurrency

Personal Projects

TiMoto AI – Backend & ML Serving Platform | *Go, Python, Django, gRPC, PostgreSQL, AWS, Terraform* tim

- **Sole engineer** — built and operate end-to-end AI platform: **Python/Django** REST backend, gRPC inter-service layer, **React** frontend, **PostgreSQL**, and AWS cloud infrastructure
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause via structured logging to circular resource acquisition; redesigned call sequencing — **100% success rate** at **sub-50ms p99**
- Selected **vLLM** with PagedAttention over naive inference to eliminate KV cache memory fragmentation under concurrent load; deployed with continuous batching — zero OOM failures at production traffic
- Architected multi-AZ **ECS Fargate** with **Terraform** IaC and circuit breaker pattern — **99.9% uptime**, **44% cost reduction** to $40–60/month; LLM cost guardrails with anomaly detection for GPU memory and inference throughput

Pulumi – Open Source Contributor | *Go, TypeScript, IaC* 24.4K★ github.com/pulumi/pulumi

- Contributed **Go** CLI features enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

Technical Skills

Languages: Go, TypeScript, Python, Java, C++, JavaScript, SQL, Rust, Bash
Web & Backend: Django, FastAPI, React, REST APIs, gRPC, Protocol Buffers, Node.js, PostgreSQL (indexing, aggregations, schema design), Redis
Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS, DynamoDB), GCP, Kubernetes, Docker, Terraform, CI/CD, GitHub Actions
Distributed Systems: Exactly-once semantics, Idempotent APIs, Circuit breakers, Lock-free data structures, Raft/Paxos, Linearizability, Fault tolerance
ML & AI: vLLM, PagedAttention, Continuous Batching, LLM cost guardrails, PyTorch, LLM-as-a-judge evaluation